

ABIYSHEK SATHIANARAYANAN

AI/ML Engineer • Full-Stack Developer • Quantum Computing

aabiyshek@gmail.com | +91 6374724269 | linkedin.com/in/abiyshek | github.com/Abiyshek | leetcode.com/u/sheky_001100 |

Puducherry, India

PORTFOLIO: Abiyshek.github.io/Abiyshek_portfolio

PROFESSIONAL SUMMARY

Results-driven AI/ML engineer and full-stack developer with 3 internships and a 9× hackathon record. I deliver immediate value by shipping end-to-end web apps in major domains of computer science, automating workflows with computer vision, reducing operational risk via intelligent diagnostics, and building platforms that scale. Community leader who has upskilled 500+ students in Generative AI.

TECHNICAL SKILLS

Languages: Python, Java, C

AI / ML: PyTorch, Scikit-learn, OpenCV, NumPy, Pandas, GNN, Generative AI, Prompt Engineering, NLP

Web & Backend: React.js, Node.js, MongoDB, Flask, MERN Stack

Cloud & Tools: AWS (S3, Bedrock, IAM), Git, GitHub, Qiskit (Quantum Computing)

CS Concepts: Data Structures, OOP, Agile/Scrum

EXPERIENCE

Toast2Host | Software Developer Intern - Web Development

May 2025 – Present

- Built full-stack features with MERN stack (MongoDB, Strapi, React.js), cutting page load time by 20% and delivering 3 product milestones on schedule.
- Integrated and optimised MySQL queries, reducing average server response time by 25% and improving overall application reliability.

Flaunch | Generative AI Intern

Sep 2024 – Oct 2024

- Fine-tuned Generative AI models and applied advanced prompt engineering, boosting output relevance by ~30%; delivered reusable templates adopted by the internal team.
- Researched and benchmarked 4+ LLM architectures, producing a comparative report that guided the team's model selection strategy for client deployments.

Rinex | AI / Machine Learning Intern

Jan 2024 – Apr 2024

- Designed and trained 5+ ML models (classification, regression, NLP); improved accuracy by 15% via hyperparameter tuning and authored technical documentation for team onboarding.
- Automated the data preprocessing pipeline using Python (NumPy, Pandas), reducing manual data preparation time by 40% and ensuring consistent model input quality.

KEY PROJECTS

EVASAFE : AI Women's Safety System | Python, OpenCV, TensorFlow, Flask, AWS

- Built a real-time CV pipeline on CCTV feeds detecting suspicious behaviours (loitering, aggression) at ~92% accuracy using TensorFlow object detection models.
- Developed a Flask microservice that triggers authority alerts within 2 seconds of threat detection; designed modular architecture for scalable city-wide AWS deployment.

ML for Brain Graphs : Autism Classifier | Python, PyTorch, NetworkX, fMRI Data

- Constructed brain-connectivity graphs from fMRI datasets and trained Graph Neural Networks (GNNs) to classify Autism Spectrum Disorder vs. neurotypical subjects with high accuracy.
- Implemented feature extraction on node/edge neural attributes and built visualisation dashboards to surface connectivity irregularities, enabling interpretable clinician insights.

ATHLATONICS : Neuro-Adaptive Sports Platform | React.js, EEG, Blockchain, IPFS, RAG

- Processed real-time EEG brainwave signals (alpha, beta, theta, gamma) from a wearable device to assess athlete cognitive states, delivering live coaching guidance via a React.js dashboard.
- Secured athlete records on blockchain with IPFS storage for tamper-proof integrity; built RAG-powered AI video analysis engine to convert match footage into actionable coach/scout narratives.

EDUCATION

B.Tech - Information Technology, Pondicherry University

2023 – 2027 (Pursuing)

Sri Manakula Vinayagar Engineering College, Puducherry **CGPA: 7.97 / 10**

Class XII - Kendriya Vidyalaya No. 2, Puducherry

2022 – 2023 | 71.8%

Class X - Kendriya Vidyalaya No. 2, Puducherry

2020 – 2021 | 77.8%

CERTIFICATIONS, ACHIEVEMENTS & LEADERSHIP

- Languages:** TAMIL (Native), ENGLISH (Advanced), HINDI (Professional), GERMAN (B1, Professional), SANSKRIT (Basics).
- Achievements:** 9× Hackathon Winner | Advanced Prompt Engineering - IBM | Generative AI Specialization - Flaunch | IBM Quantum Computing Basics | AI Model Development & Optimization.
- QBrAlnix:** Spearheaded 5+ AI/ML seminars reaching 500+ students; mentored peers on Generative AI, prompt engineering, and real-world model building.